

Binary code

Computers only understand electrical signals, where ON is represented by 1 and OFF is represented by 0. Binary code is used to translate these numbers into electrical signals.

SEE ALSO

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◀ 80–81 Bits and digitization

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Language of computers

The binary number system uses only two digits, 0 and 1. Every operation that a computer carries out and every piece of data that it stores, or processes, is represented as binary code. Binary code uses 0s and 1s to represent a letter, digit, character, or part of another item, such as music or a picture. A binary number 8 bits long can have any value from 0 to 255. This means that an 8-bit binary number could represent any one of those 256 characters.

▷ Binary translators

Humans find binary numbers difficult to interpret. So most people write programs in languages closer to human languages. These are translated into binary by programs called compilers and interpreters.



= 1111111111111111

ABC

= 01000001 01000010 01000011



= 1000 1010

123

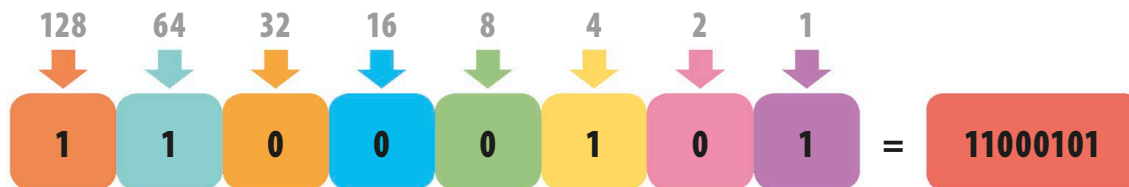
= 01111011

Binary to decimal

People usually count using the decimal system. This makes it difficult to look at a binary number and know what it represents. In a binary number, the least significant bit (LSB) is the bit furthest to the right and the most significant bit (MSB) is the bit furthest to the left. Between these two, the range of values that can be written doubles with each digit added.

▽ Conversion

The first column in the binary system is the LSB on the right and is equivalent to one. Subsequently, each column is worth double the column to its right. To get the equivalent decimal value of a binary number, multiply the value of each column with its respective binary digit and then add all the individual results.



$$1 \times 128 + 1 \times 64 + 0 \times 32 + 0 \times 16 + 0 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 =$$

197

Therefore, the binary code for 197 is 11000101.

IN DEPTH

Leibniz and the *I Ching*

The binary number system was invented by German philosopher Gottfried Leibniz (1646–1716). He highlighted its relation to the Chinese text *I Ching*, which uses arrangements of six lines to guide the user to relevant advice. The six lines are in pairs representing the concept of Yin and Yang, or, according to Leibniz, 1 and 0.

